

# E-Commerce Return Rate Reduction Analysis

Internship Project 2 | Elevate Labs | Ankita Daweshar | June 2026

## Abstract

Product returns are a major cost driver for e-commerce businesses, impacting profitability through reverse logistics, restocking, and lost sales. This project analyses 5,000 e-commerce orders to understand why customers return products and how return rates vary by product category, geography, payment method, and shipping channel. A logistic regression model was built to predict the probability of an order being returned, enabling proactive risk flagging before dispatch. The analysis revealed an overall return rate of 29%, with Clothing as the highest-risk category (37.4%) and certain cities showing return rates as high as 50%. The final deliverable is a 4-page interactive Power BI dashboard with drill-through filters, a Python prediction codebase, a high-risk orders CSV, and this summary report.

## Introduction

E-commerce companies lose significant revenue when customers return products. Every return triggers reverse shipping costs, quality inspection, restocking effort, and a missed sale. Despite this, most businesses lack visibility into which products, customer segments, or operational choices (such as shipping or payment method) are driving returns. This project addresses that gap by combining exploratory data analysis, statistical summarisation, and machine learning to identify the key drivers of returns and to build a predictive early-warning system that scores every order with a return-risk probability.

**Objective:** Identify why customers return products and how return rates vary by category, geography, and marketing channel; predict return probability per order to flag high-risk orders before dispatch.

## Tools Used

| Category         | Tools / Libraries                                                           |
|------------------|-----------------------------------------------------------------------------|
| Programming      | Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn)                   |
| Machine Learning | Logistic Regression (Scikit-learn), StandardScaler, train/test split        |
| Visualization    | Power BI Desktop (DAX measures, slicers, drill-through), Matplotlib/Seaborn |
| Database         | SQL (MySQL Workbench)                                                       |
| Development      | Visual Studio Code, Jupyter Notebook, Git & GitHub                          |

## Steps Involved

- 1. Data Collection & Cleaning** — Sourced a 5,000-row e-commerce returns dataset (23 features, zero missing values) covering order, product, customer, shipping, payment, and return details from Jan 2022 to Sep 2025.
- 2. Exploratory Data Analysis** — Computed return rate by product category, city, payment method, shipping method, gender, and year using Python (Pandas, Matplotlib, Seaborn). Generated 11 visualizations and exported a high-risk products CSV (products with return rate  $\geq 40\%$ ).
- 3. Feature Engineering & Modelling** — Encoded categorical features, scaled numeric features, and trained a Logistic Regression classifier (80/20 train-test split, class-balanced) to predict return probability for each order.
- 4. Model Evaluation** — Assessed performance using accuracy, ROC-AUC, confusion matrix, and classification report. The model achieved 71% accuracy on unseen test data.
- 5. Risk Scoring** — Applied the trained model to all 5,000 orders to generate a Return Risk Score (0–1) and categorical Risk Label (Low/Medium/High), exported as CSVs for downstream use.
- 6. Dashboard Development** — Built a 4-page interactive Power BI dashboard (Overview, Segments, Risk Score Analysis, Insights) with KPI cards, charts, slicers, drill-through filters, and a fixed navigation panel.
- 7. Reporting** — Documented findings, methodology, and business recommendations in this report, alongside a reusable Python codebase (Jupyter notebooks and a standalone script).

## Key Findings

Overall return rate across all orders is **29.0%** — nearly 1 in 3 orders is returned.

**Clothing** has the highest return rate at **37.4%**; Books has the lowest at 22.6%.

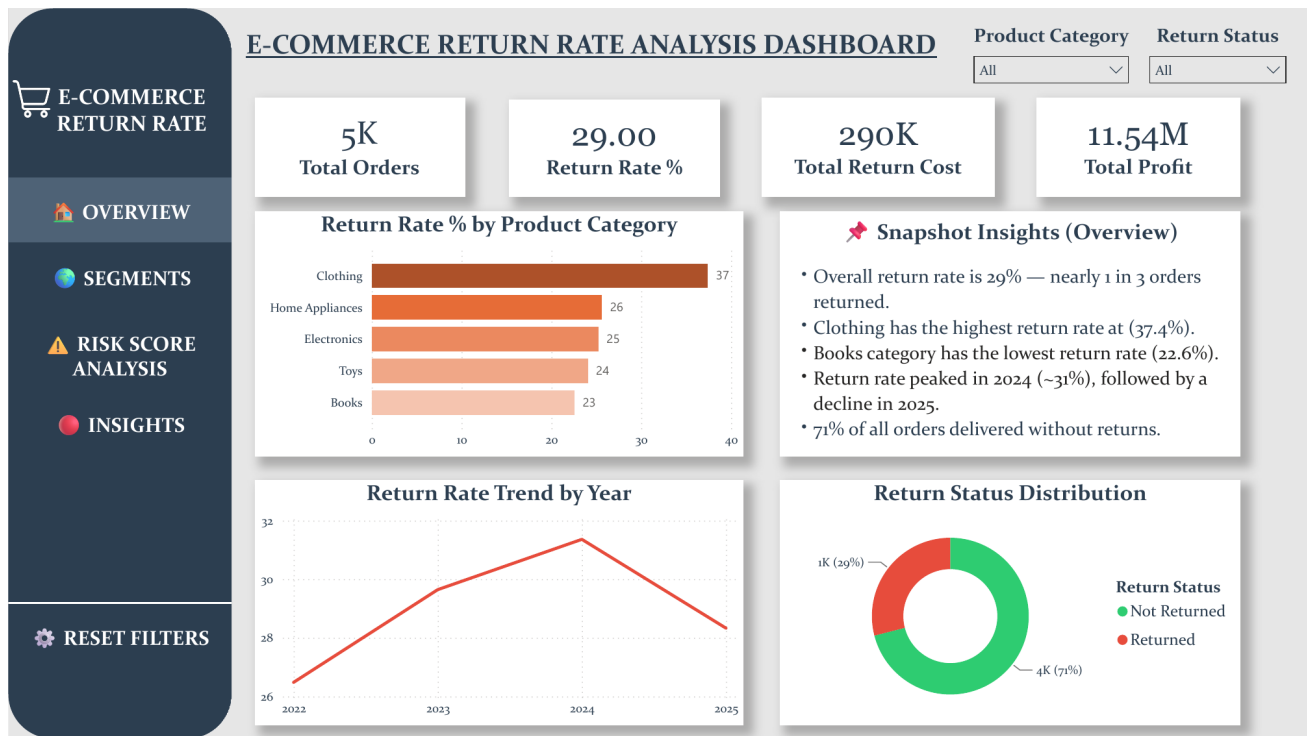
**City44** is the riskiest location with a **50% return rate**, followed by City85 (43.9%) and City77 (43.2%).

**Credit Card** payments show the highest return rate (31%), with COD close behind (30.5%).

The Logistic Regression model achieves **71% accuracy** on test data, identifying 2,406 orders as High Risk.

Total direct return cost across the dataset is approximately **Rs. 2,90,000**.

## Dashboard Snapshot



Power BI Dashboard – Overview page (Page 1 of 4)

## Business Recommendations

- Investigate Clothing category quality and sizing — the top driver of returns.
- Audit suppliers and logistics for City44, City85, and City77.
- Add order verification steps for Credit Card and COD transactions above a price threshold.
- Use the ML risk score to flag high-risk orders for quality re-checks before dispatch.

## Conclusion

This project demonstrates a complete, end-to-end analytics workflow for tackling e-commerce returns — from data cleaning and exploratory analysis through to predictive modelling and an interactive business intelligence dashboard. The findings provide actionable, data-backed direction for reducing return rates, particularly in the Clothing category and in specific high-risk cities. The predictive risk-scoring system offers a practical mechanism for the business to proactively flag and review orders before they are shipped, potentially reducing reverse-logistics costs and improving customer satisfaction.